A Maximal-Disjoint-Family Argument Refutes the Infinite-Lebesgue-Measure Independence Conjecture

Candidate counterexample packet
arXiv:2503.06217, Conjecture 4.6

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Status. Candidate full ZFC disproof, subject to expert review. In fact the argument constructs the stronger object that the source calls a $\mathfrak{c}$-separable *partition*; hence the conjectured failure of a $\mathfrak{c}$-separable envelope is impossible.

1 Source question and exact consequence

Rodríguez-Vidanes and Sampedro study Baker's infinite-dimensional Lebesgue measure μ on the Borel space $(\mathbb{R}^{\mathbb{N}}, \mathcal{B}_\infty)$ [1]. Their Conjecture 4.6, printed on page 19, reads:

> There is a model of ZFC in which $(\mathbb{R}^{\mathbb{N}}, \mu)$ does not admit a $\mathfrak{c}$-separable envelope.

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Conjecture 4.6. *There is a model of ZFC in which* $(\mathbb{R}^{\mathbb{N}}, \mu)$ *does not admit a* $\mathfrak{c}$*-separable envelope.*

Assuming this conjecture, since $(\mathbb{R}^{\mathbb{N}}, \mu)$ is relatively nonatomic and has the separability property, Theorem 4.5 yields that the statement

$$L^p(\mu) \cong \ell^p(\mathfrak{c}, L^p[0,1]) \quad (p \neq 2)$$

is independent of ZFC (i.e., neither provable nor refutable within ZFC).

The authors state that this would make

$$L^p(\mu) \cong \ell^p(\mathfrak{c}, L^p[0,1]) \qquad (p \neq 2) \tag{1}$$

independent of ZFC. We prove instead that ZFC itself gives the requisite partition and therefore proves (1), for every $1 \leq p < \infty$.

2 Definitions and source inputs

In the terminology of [1, Definition 4.2], a family $\{X_i : i \in I\}$ is a κ-separable partition if $|I| = \kappa$, every X_i is a nonzero, sigma-finite, separable measurable set, distinct members intersect in a null set, and

$$\nu(E) = \sum_{i \in I} \nu(E \cap X_i) \tag{2}$$

for every measurable E of finite measure. The sum over an arbitrary index set means the supremum of its finite subsums. A separable envelope only asks for (2) when E is contained in the union of the X_i; thus a separable partition is stronger.

We use three facts proved in the source:

1. $\mathcal{B}_\infty$ is generated by countably many rational cylinder sets, and every finite-measure measurable set is separable [1, Lemma 2.3];

2. Section 3 constructs $\mathfrak{c}$ many measurable rectangles C_A, each of measure one, with $\mu(C_A \cap C_B) = 0$ for $A \neq B$ [1, proof of the main theorem of Section 3];

3. $(\mathbb{R}^{\mathbb{N}}, \mu)$ is relatively nonatomic [1, Appendix A], and a relatively nonatomic space with a κ-separable partition has $L^p(\nu) \cong \ell^p(\kappa, L^p[0,1])$ [1, Theorem 4.3].

3 Proof intuition

The only apparent difficulty is that an uncountable almost-disjoint family need not have a measurable union. Maximality avoids taking that union. Fix a finite-measure test set E. Among a pairwise-null family, only countably many members can capture positive measure from E. Remove precisely those countably many pieces. If a positive remainder survived, it would itself be a finite-measure set almost disjoint from every member of the maximal family, contradicting maximality. This proves the full partition identity locally on each finite E.

The source's continuum-sized rectangle family supplies the lower cardinal bound. Countable generation of the ambient sigma-algebra supplies the upper bound. No form of CH is used.

4 The maximal-disjointization lemma

Lemma 1. *Let (X, Σ, ν) be a measure space, and let $\mathcal{X}$ be a maximal family of measurable sets such that*

$$0 < \nu(X_i) < \infty \quad (X_i \in \mathcal{X}), \qquad \nu(X_i \cap X_j) = 0 \quad (X_i \neq X_j).$$

Then for every $E \in \Sigma$ with $\nu(E) < \infty$,

$$\nu(E) = \sum_{X_i \in \mathcal{X}} \nu(E \cap X_i). \tag{3}$$

Proof. For every finite $F \subset \mathcal{X}$, pairwise nullity gives

$$\sum_{X_i \in F} \nu(E \cap X_i) \leq \nu(E).$$

Let

$$J_E := \{X_i \in \mathcal{X} : \nu(E \cap X_i) > 0\}.$$

This set is countable. Indeed, for every $n \geq 1$ there can be only finitely many i with $\nu(E \cap X_i) \geq 1/n$; otherwise finite subsums would exceed $\nu(E)$. Every positive number is at least $1/n$ for some n, so J_E is a countable union of finite sets.

Enumerate J_E (with the finite case included) and use countable additivity after discarding the pairwise-null overlaps. Then

$$\nu\left(E \cap \bigcup_{X_i \in J_E} X_i\right) = \sum_{X_i \in J_E} \nu(E \cap X_i).$$

If the right-hand side were strictly smaller than $\nu(E)$, the measurable set

$$F_E := E \setminus \bigcup_{X_i \in J_E} X_i$$

would satisfy $0 < \nu(F_E) < \infty$. It is disjoint from each member of J_E. For $X_i \notin J_E$, we have $\nu(F_E \cap X_i) \le \nu(E \cap X_i) = 0$. Thus F_E could be adjoined to $\mathcal{X}$, contradicting maximality. Hence equality holds, which is (3), since all terms outside J_E vanish. □

5 Main result

Theorem 2. *ZFC proves that $(\mathbb{R}^{\mathbb{N}}, \mathcal{B}_\infty, \mu)$ admits a $\mathfrak{c}$-separable partition. Consequently Conjecture 4.6 of [1] is false, and*

$$L^p(\mu) \cong \ell^p(\mathfrak{c}, L^p[0,1])$$

isometrically for every $1 \le p < \infty$, without CH.

Proof. Let $\mathcal{A}_0 = \{C_A : A \in \mathcal{A}\}$ be the source's continuum-sized family of measure-one rectangles with pairwise null intersections. Order by inclusion the families of finite positive measurable sets which contain $\mathcal{A}_0$ and whose distinct members have null intersection. The union of every chain is again such a family, so Zorn's lemma gives a maximal extension $\mathcal{X}$.

Every member of $\mathcal{X}$ has finite positive measure, hence is sigma-finite; by source Lemma 2.3 it is separable. Lemma 1 gives identity (2) for every finite-measure E. Thus $\mathcal{X}$ is a separable partition apart from checking its cardinal.

We have $|\mathcal{X}| \ge |\mathcal{A}_0| = \mathfrak{c}$. On the other hand, the countably generated sigma-algebra $\mathcal{B}_\infty$ has cardinal at most $2^{\aleph_0} = \mathfrak{c}$, so $|\mathcal{X}| \le \mathfrak{c}$. Hence $|\mathcal{X}| = \mathfrak{c}$ in ZFC. This proves the first assertion. The isometry follows immediately from relative nonatomicity and source Theorem 4.3. □

6 Verification, novelty, and review focus

The proof was audited in five independent places: existence of the maximal extension; countability of J_E; measurability and measure of the remainder; admissibility of each maximal-family member; and the exact cardinal $|\mathcal{X}| = \mathfrak{c}$. Notice especially that no uncountable union is asserted to be measurable. The argument proves the source's stronger partition definition, so no ambiguity in its weaker envelope definition affects the result.

A bounded novelty search through 11 August 2026 covered the run's solution, attempt, proof-gap, and registry indexes; the exact wording of Conjecture 4.6; the title and arXiv id; the DOI; author names; and combinations of "separable envelope," "maximal disjoint family," and "ZFC." The current arXiv PDF and published metadata were also checked. No correction, erratum, or later resolution was found. This is evidence about retrieval, not proof of novelty.

Human-review recommendation. High priority. The conclusion is unexpectedly stronger than the source's CH theorem, while the proof is short. Review should focus on whether "separable subset" in Definitions 4.2 and 4.4 is exactly the separability property supplied by Lemma 2.3, and on the remainder step in Lemma 1. Under the source's printed definitions, both checks are direct.

References

[1] D. L. Rodríguez-Vidanes and J. C. Sampedro, *Isometric classification of the L^p-spaces of infinite dimensional Lebesgue measure*, arXiv:2503.06217; Banach J. Math. Anal. **20** (2026), article 7, DOI: 10.1007/s43037-025-00473-y.